Identification of an Unknown- “A”

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Introduction

Spectroscopy data was given for the unknown compound A. The structure of A was then identified using this data. This is necessary since it is often much safer, cheaper, and more expedient to run a spectroscopy test on an unknown compound than to try other identification methods. Volatile and ignitable compounds are both hard to test with the Mel-Temp since it requires high temperatures, and chemical identification tests can be incredibly dangerous since a truly unknown substance might react violently with one of the chemicals used to test it.

NMR

Nuclear Magnetic Resonance, or NMR, spectroscopy uses the magnetic spin states of the molecules to generate data which helps chemists identify the molecule. This is usually done with hydrogen but is also performed on carbon-13, and isotopes of other molecules with odd numbers of neutrons. However, since all materials have diamagnetic properties, any substance can be analyzed (with varying degrees of efficacy) utilizing NMR spectroscopy. The induced magnetic field interacts with the magnetic field induced by the region of electron density, creating an effect known as electron shielding. This effect occurs at different extents depending on the electron density around a particular proton. Protons with less electron density are considered deshielded. Due to this shielding differential, protons absorb different frequencies of rf radiation, enabling the detection of different signals. Modern NMR machines produce this rf radiation through a technique known as pulsed Fourier transform NMR. A broad spectrum of electromagnetic radiation is emitted simultaneously, then the decay as excited nuclei return to normal is measured. This sine curve (known as a free induction decay, or FID) is then evaluated mathematically through a Fourier transform to convert it into a set of signals. Compounds cause higher or lower peaks due to constructive and destructive interference between the waves being emitted. Most spectrometers use incredibly cold temperatures generated by specialized chambers to operate superconductors which generate massive magnetic fields. Each FID can be acquired in 1-2 seconds, meaning that a multitude of readings can be taken in a very short period of time and averaged to produce a more accurate result. These signals can then be compared to signals from known compounds and structures to glean information about the unknown in consideration.

FTIR

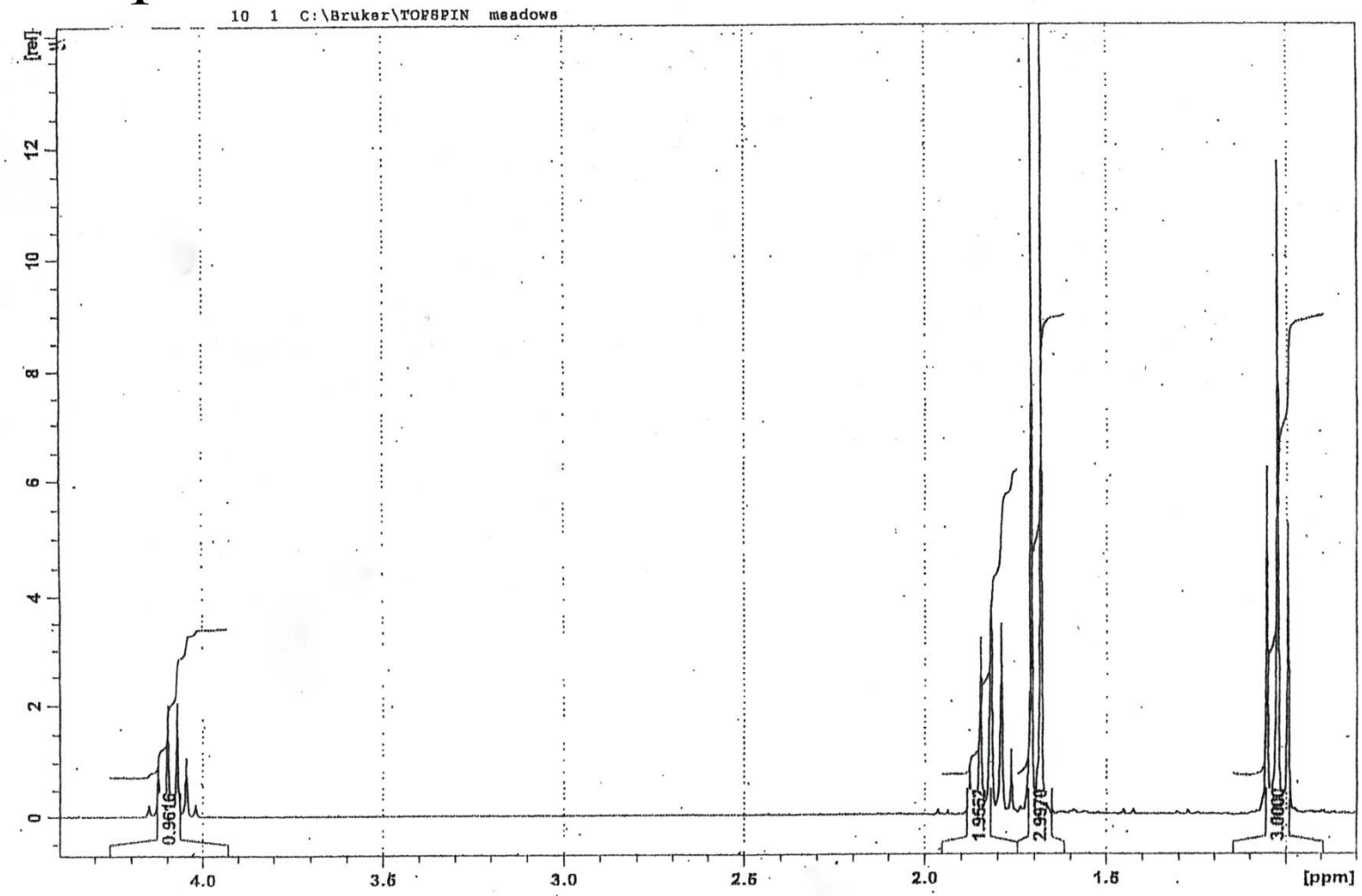
Infrared, or IR, spectroscopy is another method used to help identify unknown compounds. Infrared energy is passed through the compound. Since the energy states of bonds are quantized and can only attain certain values, this produces characteristic absorption bands. These bands can be differentiated and correlated to different types of chemical bonds within the molecule. They are produced due to the fact that these different bonds absorb different amounts of infrared energy, causing them to vibrate more. Infrared spectroscopy is also known as vibrational spectroscopy for this reason. This energy is not enough to break the bonds, but the increased vibrations cause the sample to heat up. The IR spectrometer detects which frequencies made it through the molecule, then that information is used to deduce which ones were absorbed. Signals used to be reported in wavelength (micrometers) but are currently most often reported in terms of wavenumber (frequency). Wavenumber plots are typically easier to read than wavelength plots. Higher wavenumbers correlate to stronger bonds, meaning triple bonds and bonds to hydrogen have the highest wavenumbers, while double bonds and other single bonds have lower wavenumbers. IR spectra are divided into two regions: the diagnostic region and the fingerprint region. The diagnostic region is a result of the two higher energy types of bonds and is easier to analyze. The fingerprint region is a result of the two lower energy types of bonds, and its name comes from the fact that each compound has a unique pattern of signals that are much more difficult to analyze. Bonds that are weakened by resonance produce a lower wavenumber than non-resonance stabilized counterparts, despite the molecule having more double bonds.

GCMS

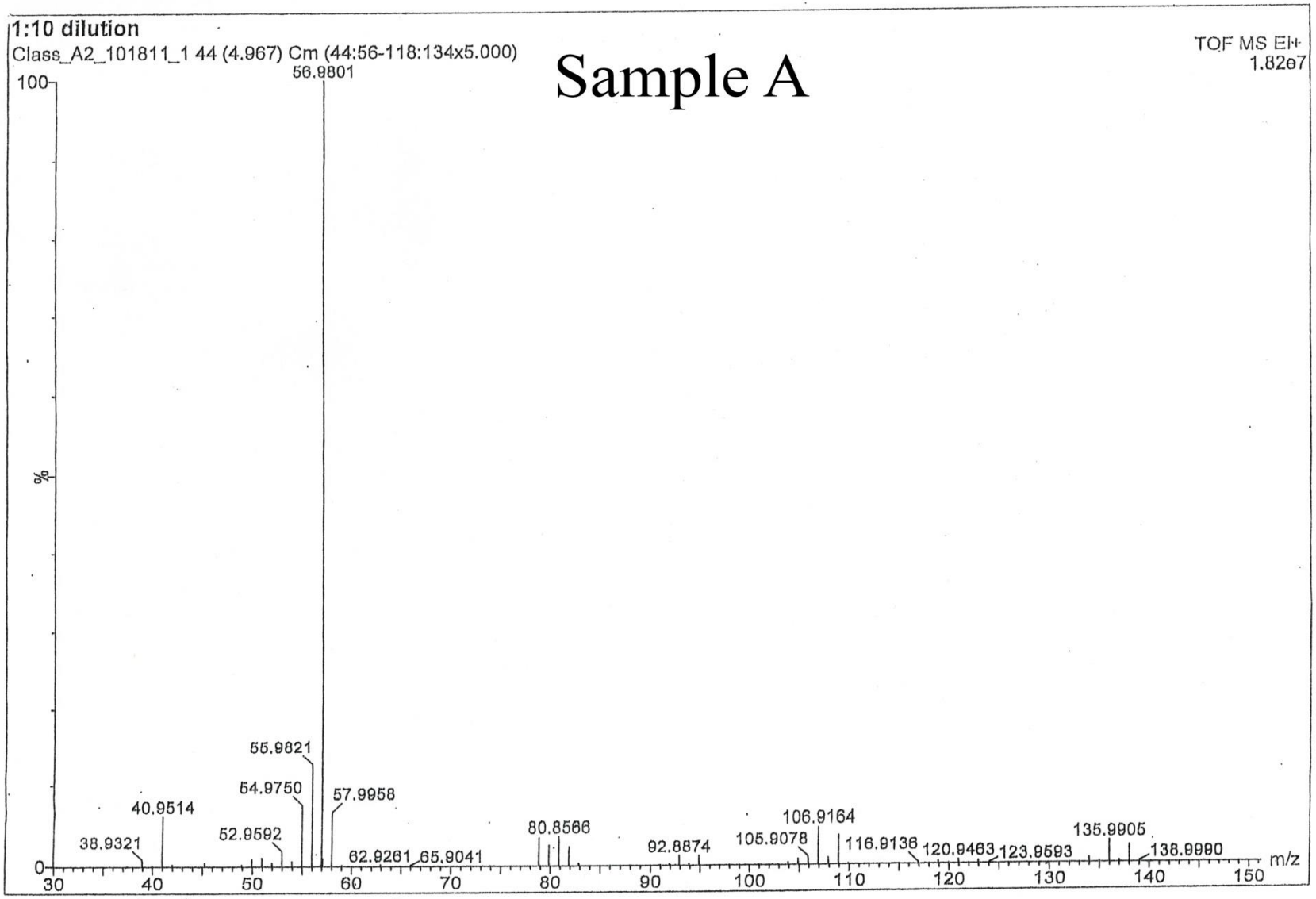
Gas chromatograph-mass spectrometers, or GCMS, are hybrid instruments capable of analyzing mixtures of compounds. They provide information on the number of components, relative amounts of each, and possible identification. Gas chromatography requires volatile components, which are vaporized and then ionized. Ionization is often achieved by bombarding the compounds with incredibly high energy electrons, which eject electrons from the molecules to create ions. These ions are radical cations, which are prone to fragmentation. Typically, one fragment is the cation, while the other carries the radical electron. The ionized particles are then accelerated into a magnetic field. The gas chromatograph-mass spectrometer uses a quadrupole filter, named for the four parallel stainless-steel rods which guide resonant ions to the detector. Each pair of rods is charged with opposing DC voltage, which is superimposed with high frequency AC voltage. The combined fields cause molecular oscillation, which throws non-resonant molecules out of the path, but allows molecules of specific mass to charge (m/z) ratios that resonate with the field to pass through and hit the detector. Mass spectrometers work by utilizing the different trajectories of ions with differing weights through the field. These ions hit the detector in predictable locations, causing peaks in the spectra that can be identified. The highest peak is called the base peak, and the heights of other peaks are described relative to the base peak. If the molecule does not fragment, the base peak is the molecular weight of the compound. The molecular weight can be determined by the x-axis of the graph of the spectra, which is the m/z ratio. This and other rules for specific atoms, such as the nitrogen rule, can help identify the composition of the compound.

NMR Sample A

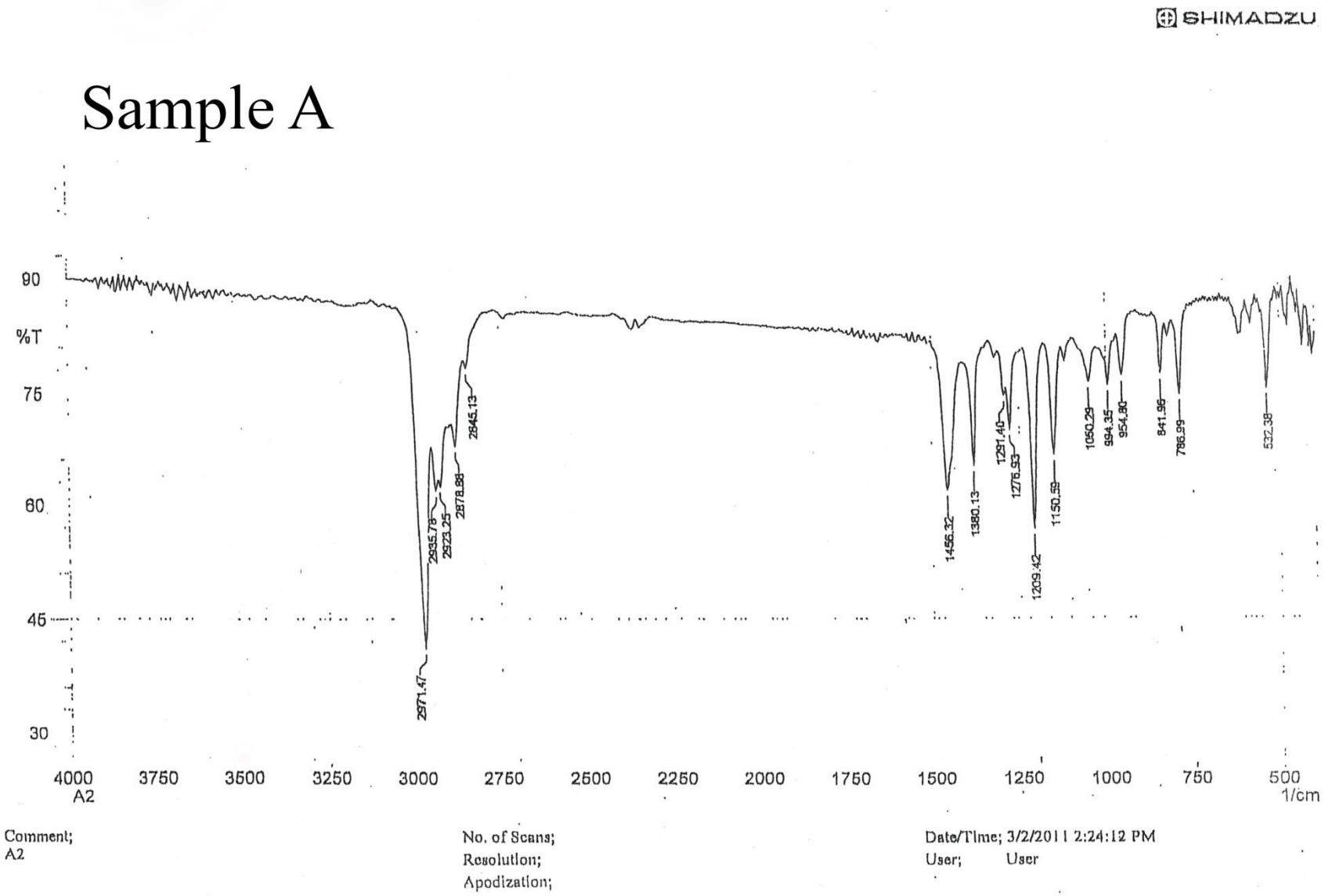
Sample A



GCMS Sample A



FTIR Sample A



Conclusions

From the sample data, 2-bromobutane is the most likely candidate for the unknown. From the NMR spectra, four signal groups were identified. These were broken into a sextet (0.9616), quintet (1.9557), doublet (2.9970), and triplet (3.0000), with respective integration factors. Rounding the integration factors led to a ratio of 1:2:3:3, which gives 9 hydrogens. The protons of 2-bromobutane fit these characteristics: there are four groups of protons, one with one neighbor, one with five neighbors, one with four neighbors, and one with two neighbors, leading to the observed multiplicities. This molecule is chiral, however it does not change the multiplicity since the chiral carbon is bonded to the bromine and a single hydrogen. From the IR spectra, no peaks were observed in the 2300-2100 and 1850-1620 ranges, meaning there were likely no double or triple bonds to carbon. Peaks were observed above these ranges in the diagnostic region, as well as below in the fingerprint region, likely corresponding to C-H and C-C bonds, respectively. From the GC-MS spectra, the top peak has a value of 56.9801. When 2-bromobutane is hit with electrons, the most likely outcome is fragmentation with the loss of the bromine. The molecular weight of 2-bromobutane is around 136. Subtracting the bromine at 79, a value of 57 is obtained, which is almost exactly the value of the top peak in the GC-MS data.

References

Keller L, Joseph K, and Miguel S. 2023 Fall. CHEM 2081 Organic Chemistry 2 Lab Manual [lab manual]. Auburn (AL): Auburn University.

Klein, David. Organic Chemistry, 4th ed.; John Wiley and Sons, Inc., 2021